

Abhishek V. Joshi

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241 Loomis Laboratory, 1110 W Green St, Urbana, Illinois

EDUCATION

PhD, Physics, University of Illinois Urbana-Champaign	2019–present
Advisor: Professor Charles Gammie	GPA: 3.9/4.0
BS, Engineering Physics, University of Illinois Urbana-Champaign	2015–2018
Summa Cum Laude, Computer Science Minor	GPA: 3.83/4.0

RESEARCH INTERESTS

Black hole accretion, general relativistic radiative transfer (GRRT), general relativistic and non-relativistic magnetohydrodynamics (GRMHD, MHD), numerical relativity, astrophysical plasmas, High-Performance Computing (HPC), High-Throughput Computing (HTC)

WORK EXPERIENCE

Graduate Research Assistant Intern, Los Alamos National Lab	2025
Project: jets of embedded stellar mass black holes in AGN disks.	
Graduate Research Assistant, UIUC	2020–2024
Projects: circular polarization of simulated black hole images, effects of anisotropic electrons on spectra of supermassive black holes, Event Horizon Telescope Collaboration projects.	
Graduate Teaching Assistant, UIUC	2019–2020, Fall 2022, Fall 2024, Spring 2025
Software Intern at Thermo Fisher Scientific San Jose, CA	Summer 2019
Project: developed methods for real-time optimization of ion source inlet sprays for mass spectrometers (Patent US11791144B2).	

RESEARCH EXPERIENCE

Radiative Transfer methods

Developed and maintained deterministic and Monte Carlo radiative transfer codes in the AFD-Illinois group.
Experience with polarized modeling, circular polarization in particular.
Generated large libraries of simulated black hole images and applied statistical methods for model inference with observations for both individual and Event Horizon Telescope collaboration (EHT) projects.

Magnetohydrodynamic Modelling of Accretion Flows

Generated and analyzed simulations of general relativistic accretion near supermassive black holes.
Computed emission proxies and jet power statistics for model inference.
Generated simulations of AGN disks, modeling feedback from embedded stellar mass black holes.

Numerical Relativity

Generated simulations of eccentric spin-aligned binary black holes using the Einstein Toolkit. Compared numerical waveforms to template library waveforms to investigate effects of high-order modes.

AWARDS AND HONORS

Mavis Future Faculty Fellowship	2025–2026
Scott Anderson Outstanding Graduate Assistant Award	Spring 2025
University Fellowship Award for Research Accomplishments	Fall 2023

Gregory, J.M & L.C. scholarship	2017–2018
UIUC James Scholar Honors	2016–2018

SELECTED PUBLICATIONS

Lists: NASA ADS (recommended), Google Scholar, ORCID. 1304 citations, 2 first-author

COMPUTER SKILLS

Programming languages: C, C++, Python, shell-scripting, Lua, Java

Scientific software: numpy, scipy, Mathematica, pytorch, matplotlib, yt, Paraview

Version control and Containerization: git, svn, Docker, Singularity

Parallelization frameworks: openMP, MPI, HTCondor, CUDA

MENTORING

Illinois Guidance for Physics Students (GPS)	2023–present
Mentored three undergraduate students about physics research, grad school applications and general academic guidance.	

Computational Astrophysics Group Mentoring	2021–present
Mentored four undergraduate students on their own projects within Charles Gammie’s research group.	

UIUC Young Scholars Program	Summer 2018
Mentored a high school student over the summer on a simulation visualization project.	

TEACHING

Teaching Assistant, University of Illinois Urbana-Champaign

PHYS 598, Computational Physics and Astrophysics	Fall 2022, Fall 2024
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PHYS 214, University Physics: Quantum Physics	Summer 2020
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PHYS 212, University Physics: Electricity & Magnetism	Spring 2020
Ranked excellent teacher by students.	

PHYS 102, College Physics: E&M and Modern Physics	Fall 2019, Spring 2025
Ranked excellent teacher by students.	

OUTREACH AND VOLUNTEERING

Education Justice Project (EJP), Writing and Math Program Tutor	2023–present
Tutored students enrolled within the EJP initiative at the Danville Correctional Center in math, writing and general black hole astrophysics concepts.	

Speaker, Satkama High School, Hyderabad, India	Jan 2024
Engaged with students from 6th to 10th grades by presenting concepts of horizon-scale black hole astrophysics with demos and animations.	

Lab Presenter, Conference for Undergraduate Women in Physics (CUWiP)	Jan 2023
Gave a talk to participants of the conference about Charles Gammie’s research group activities and EHT science.	

Black hole science demo, University Primary School	May 2022
Interacted with students from 2nd to 5th grades about black holes and EHT science. Coordinated an outdoor activity for the students.	